



MCT5210 – Control Engineering III

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Transfer Functions VS State Space

- Consider as system with the transfer function

$$G(s) = \frac{Y(s)}{U(s)}$$

- The transfer function is defined as the ratio of the Laplace transform of the output to the ratio of the Laplace transform of the input.

The system may be represented in state space as

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

where x is the state vector, u is the input and y is the output. The Laplace transform of the above equation is

$$sX(s) - x(0) = AX(s) + BU(s)$$

$$Y(s) = CX(s) + DU(s)$$

Therefore at zero initial conditions

$$sX(s) - AX(s) = BU(s)$$

or

$$(sI - A)X(s) = BU(s)$$



Transfer Functions VS State Space

or

$$X(s) = (sI - A)^{-1}BU(s)$$

substituting into the output equation gives the **state transition matrix**

$$Y(s) = [C(sI - A)^{-1}B + D]U(s)$$

- By comparing with the transfer function equation, it can be seen that
- $G(s) = C(sI - A)^{-1}B + D$
- The above is the transfer function of the system in terms of A, B, C and D and can also be expressed as

$$G(s) = \frac{Q(s)}{|sI - A|}$$

where Q(s) is a polynomial in s

- It can be seen that $|sI - A|$ is equal to the **characteristic polynomial** of G(s) i.e. the eigenvalues of A are identical to the poles of G(s).



Transfer Functions VS State Space

Example: Find the characteristic equation of the state space model described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 4 \end{bmatrix} u$$

and

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + 0u$$

The characteristic equation is given by

$$\begin{aligned} G_c(s) &= \det(sI - A) \\ &= \left| s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -4 & -2 \end{bmatrix} \right| \\ &= \left| \begin{bmatrix} s & -1 \\ 4 & s+2 \end{bmatrix} \right| \\ &= s(s+2) + 4 \end{aligned}$$

Thus

$$G_c(s) = s^2 + 2s + 4$$

State Space Representation of Linear ODEs

Consider the following n^{th} order system

$$\frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + b_{m-1} \frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots + b_0 u(t)$$

Without loss of generality, assume $\mathbf{m} = \mathbf{n}$ and for simplicity take a case for which all $\mathbf{b}$ terms except $\mathbf{b}_0$ are **zero**.

Also take $y(t), \frac{dy(t)}{dt} \cdots \frac{d^{n-1} y(t)}{dt^{n-1}}$ as a set of $\mathbf{n}$ state variables.

By defining

The ODE above can be written as

$$\begin{aligned} x_1 &= y \\ x_2 &= \frac{dy(t)}{dt} \\ &\vdots \\ x_n &= \frac{d^{n-1} y(t)}{dt^{n-1}} \end{aligned}$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \\ \dot{x}_n &= -a_0 x_1 - a_1 x_2 - \cdots - a_{n-1} x_n + b_0 u \end{aligned}$$



State Space Representation of Linear ODEs

Or

$$\dot{x} = Ax + Bu$$

where

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_o & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ b_o \end{bmatrix}$$



State Space Representation of Linear DEs

The output can be expressed as

$$y = [1 \quad 0 \quad \dots \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} \text{ or } y = Cx$$

where $C = [1 \quad 0 \quad \dots \quad 0]$

Note that the state space representation of the transfer function

$$\frac{Y(s)}{U(s)} = \frac{b_n s^n + b_{n-1} s^{n-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

can also be expressed as above.



State Space Equations

Example: Express the transfer function below in state space

$$G(s) = \frac{s^2 + 7s + 2}{s^3 + 9s^2 + 26s + 24}$$

$$\frac{Y(s)}{U(s)} = \frac{\frac{1}{s} + \frac{7}{s^2} + \frac{2}{s^3}}{1 + \frac{9}{s} + \frac{26}{s^2} + \frac{24}{s^3}}$$

$$Y(s) \left(1 + \frac{9}{s} + \frac{26}{s^2} + \frac{24}{s^3} \right) = U(s) \left(\frac{1}{s} + \frac{7}{s^2} + \frac{2}{s^3} \right)$$

$$Y(s) = \frac{1}{s} \left\{ (-9Y(s) + U(s)) + \frac{1}{s} \left[-26Y(s) + 7U(s) + \frac{1}{s} (-24Y(s) + 2U(s)) \right] \right\}$$



State-Space Equations

$$x_1(t) = y(t)$$

$$\dot{x}_1(t) = -9x_1(t) + x_2(t) + u(t)$$

$$\dot{x}_2(t) = -26x_1(t) + x_3(t) + 7u(t)$$

$$\dot{x}_3(t) = -24x_1(t) + 2u(t)$$

Therefore:

$$\dot{\mathbf{x}} = \begin{bmatrix} -9 & 1 & 0 \\ -26 & 0 & 1 \\ -24 & 0 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix} \mathbf{u}(t)$$

$$\mathbf{y}(t) = [1 \quad 0 \quad 0] \mathbf{x}(t)$$

- The representation above is known as the **observable canonical** form.
- The observable canonical form arranges the coefficients of the transfer function denominator along **one column** of the A matrix.



State-Space Equations

- The system can also be represented in numerous other canonical forms.
- For example, by inspection, the **controllable canonical** form is:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [2 \quad 7 \quad 1], \quad D = 0$$

- The controllable canonical form arranges the coefficients of the transfer function denominator across **one row** of the A matrix.
- **MATLAB** uses a **variant** of the **controllable** canonical form to represent state space models when converting from transfer functions.
- For example, the following MATLAB commands

```
>> num = [1 7 2];  
>> den = [1 9 26 24];  
>> [A, B, C, D] = tf2ss(num, den)
```

produce the state space representation

$$\begin{array}{cccc} A = & & B = & C = & D = \\ & -9 & -26 & -24 & & & & & \\ & 1 & 0 & 0 & & & & & \\ & 0 & 1 & 0 & & 1 & 7 & 2 & 0 \end{array}$$



Controllability and Observability

Two issues of importance in state-space control are referred to as ***controllability*** and ***observability***.

Controllability

- A system is said to be completely state controllable if it is possible for the control system to move the state vector from any initial value to any other value in a finite time.
- This implies that it must be possible to move all the open-loop poles, by state feedback, to any closed loop location.
- The standard **test** to determine whether a system is **controllable** or not is to find the **rank** of the following partitioned matrix.

$$[\mathbf{B} \ \mathbf{A}\mathbf{B} \ \mathbf{A}^2\mathbf{B} \ \dots \ \mathbf{A}^{n-1}\mathbf{B}]$$

- If the rank of this matrix is equal to the number of rows in $\mathbf{B}$ which is equal to the number of states n then the system is completely state controllable.
- For a given square matrix to be **full rank**, then its **determinant** must be **non zero**.



Controllability

Example: The state matrices of an antennae positioning system are given below. Determine if the system is controllable.

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -5 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix}$$

Construct the following matrix:

$$[B \quad AB \quad A^2B]$$

$$A \times B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -5 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = AB = \begin{bmatrix} 0 \\ 5 \\ -25 \end{bmatrix}$$

$$A^2B = A \times AB = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -5 \end{bmatrix} \times \begin{bmatrix} 0 \\ 5 \\ -25 \end{bmatrix} = \begin{bmatrix} 5 \\ -30 \\ 125 \end{bmatrix}$$

Thus

$$[B \quad AB \quad A^2B] = \begin{bmatrix} 0 & 0 & 5 \\ 0 & 5 & -30 \\ 5 & -25 & 125 \end{bmatrix}$$

The determinant is -125 and since this is **non-zero** the system is **controllable**.



Observability

- A system is said to be completely **observable** if it is possible to **reconstruct** the **state vector** completely from **measurements** made at the system's **output**.
- For example, considering the position of a car, then it will be completely observable if by measuring speed, distance and steering wheel position, it is possible to determine where the car was parked before being driven.
- The check for observability is very similar to that for controllability. If the following derived matrix is **full rank** then the system is **observable**:

$$\begin{bmatrix} C^T & A^T C^T & (A^T)^2 C^T & \dots & (A^T)^{n-1} C^T \end{bmatrix}$$



Observability

Example: Given that a system with

$$A = \begin{bmatrix} -3 & -4 \\ -1 & 0 \end{bmatrix}, B = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, C = [-1 \quad -1],$$

The observability matrix is constructed as

$$\begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 4 & 4 \end{bmatrix}$$

The determinant of the observability matrix is zero hence the system is **not** fully **observable**.